

What phase warping buys the Figure-1 canceller non-causal and one-beat-delay, with and without warp

Working note for the CFA ear-EEG paper

September 6, 2026

What was done. Figure 1 of the paper is produced by `ekg_ilc_rt.ilc_clean(mode="noncausal")`, whose template is indexed by *fixed lag* from the R-peak. That function now takes a warp exponent γ : the template bin for a sample at offset p from beat k is $\text{ra} + \text{round}(p/w_k)$ with $w_k = (T_k/\bar{T})^\gamma$, so $\gamma = 0$ is the original code path—verified bit-identical—and $\gamma = 1$ stretches the template to each beat, i.e. normalized cardiac phase. All four combinations of $\{\text{non-causal, one-beat delay}\} \times \{\gamma = 0, \gamma = 1\}$ were run, plus $\gamma = 1/2$.

The short answer. It depends entirely on whether the artifact is time-locked to the R-peak or rides on cardiac phase, and nothing else. On synthetic data where that is controlled, warping costs up to 2.7 dB in the first regime and buys up to 1.9 dB in the second, in both cases growing with heart-rate variability. On the real recordings the sign is not uniform, and it is predictable: the three records whose R-locked energy sits in the QRS lobe lose a little from warping, while the one record whose energy sits late gains clearly. Figure 1 is drawn on a lobe-dominated record, so it loses nothing by omitting the warp. One beat of latency costs about 2.0 dB against the batch average, and that cost is essentially independent of γ : the two design choices are separable.

1 Why the comparison needs care on real data

There is no ground truth on the real recordings, and the two obvious ground-truth-free metrics are both traps at $\gamma = 0$.

Subtracting a non-causal template from the very beats that built it drives the R-triggered average of the residual to *exactly* zero, because the template *is* that average. Scored that way the fixed-lag arm reads -200 dB and the comparison is meaningless. Splitting the beats in half—template from one half, applied to the other—fixes the estimator but not the metric: the R-triggered average of the residual over *all* beats is still identically zero, since the two half-templates average back to the full one.

We therefore (i) hold out properly, partitioning samples by their nearest beat so that every sample is cleaned by a template that excludes its own beat, and (ii) score by variance reduction inside the beat window rather than by R-triggered energy. Variance reduction is coordinate-free, so it privileges neither γ , and both arms have the same number of template bins, so the comparison is not confounded by degrees of freedom either. The one-beat-delay arm needs no split: it subtracts before it updates, so each bin is already scored against beats it has not seen.

2 Synthetic, with ground truth

The shipped generator `gen_eeg` places a *fixed-shape* PQRST at every beat, so its artifact is strictly time-locked and $\gamma = 1$ cannot help by construction. Reporting only that would be rigging the answer, so we also run a stretched variant in which each beat’s waveform is scaled in time by $T_k/\bar{T}$, where $\gamma = 1$ is correct by construction. The two bracket reality.

Table 1: Synthetic sweep with ground truth: SNR improvement over the contaminated input (dB), measured against the true neural signal, mean of three seeds at 180 s each. Δ is $\gamma=1$ minus $\gamma=0$.

artifact	mode	CV_{RR}	$\gamma=0$	$\gamma=1$	Δ
time-locked	non-causal	3.0%	+12.84	+12.45	-0.39
	one-beat delay	3.0%	+10.17	+9.90	-0.28
	non-causal	5.8%	+12.05	+10.83	-1.22
	one-beat delay	5.8%	+9.59	+8.80	-0.79
	non-causal	14.3%	+9.38	+6.72	-2.66
	one-beat delay	14.3%	+7.93	+5.71	-2.23
phase-stretched	non-causal	3.0%	+12.11	+12.48	+0.36
	one-beat delay	3.0%	+9.74	+9.92	+0.18
	non-causal	5.8%	+10.96	+11.83	+0.88
	one-beat delay	5.8%	+8.90	+9.43	+0.53
	non-causal	14.3%	+6.92	+8.85	+1.94
	one-beat delay	14.3%	+5.93	+7.38	+1.45

3 Real recordings

Table 2: Real recordings, held out: variance reduction inside the beat window (dB). Main lobe is -50 to +150 ms, late is +200 to +500 ms. Δ is $\gamma=1$ minus $\gamma=0$; negative means the warp removes less.

record	mode	band	$\gamma=0$	$\gamma=1$	Δ
sub-070/run-20	non-causal	lobe	+4.37	+4.18	-0.19
	non-causal	late	+0.78	+0.71	-0.06
	one-beat delay	lobe	+4.09	+3.95	-0.14
	one-beat delay	late	+0.63	+0.58	-0.05
sub-001/run-01	non-causal	lobe	+2.07	+2.07	+0.00
	non-causal	late	+0.13	+0.12	-0.01
	one-beat delay	lobe	+1.89	+1.89	-0.00
	one-beat delay	late	+0.01	+0.01	-0.00
sub-070/run-01	non-causal	lobe	+2.19	+2.16	-0.03
	non-causal	late	+0.74	+0.66	-0.09
	one-beat delay	lobe	+2.04	+2.01	-0.03
	one-beat delay	late	+0.63	+0.59	-0.04
sub-118/run-02	non-causal	lobe	+0.06	+0.07	+0.00
	non-causal	late	+0.75	+1.34	+0.59
	one-beat delay	lobe	-0.09	-0.08	+0.00
	one-beat delay	late	+0.95	+1.45	+0.49

4 Reading

- **γ is not a free win.** Warping a template that was already aligned actively hurts, and the damage grows with the very RR variability that motivated warping in the first place.
- **The sign is predictable from where the artifact sits.** Stretching by $T_k/\bar{T}$ displaces bin o by $(o - \text{ra})(T_k/\bar{T} - 1)$ samples: essentially zero at the R-peak, large late in the cycle. So

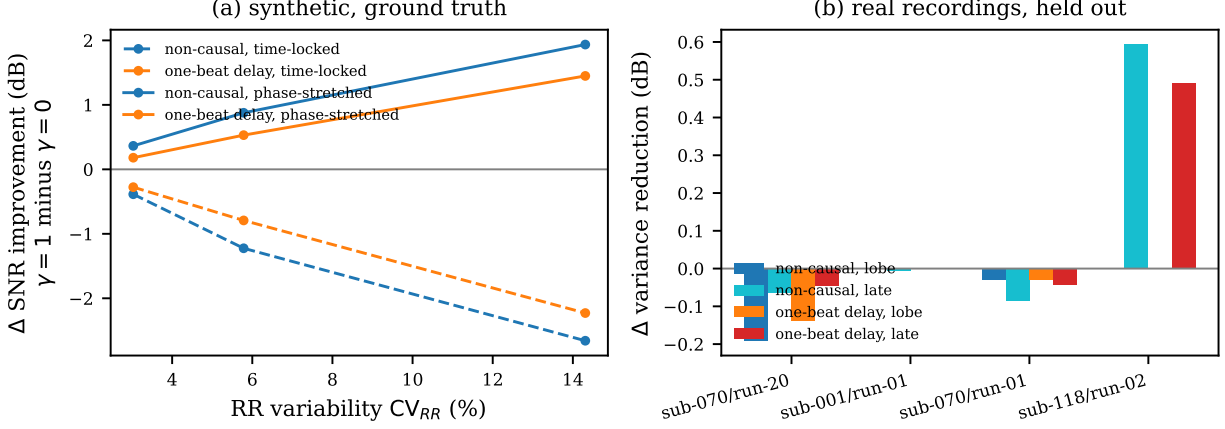


Figure 1: (a) Synthetic: advantage of $\gamma = 1$ over $\gamma = 0$ against RR variability, for both artifact regimes and both cancellation modes. The sign is set by the regime; the magnitude grows with variability. (b) Real recordings: the same difference, held out, per record and band. The lobe-dominated records lose slightly; the late-dominated record (sub-118) gains in its late band.

warping cannot help a sharp time-locked QRS deflection and can only blur it, while it realigns a genuinely phase-locked late component. Empirically the late/lobe energy ratio predicts the sign, and that ratio is measurable *before* choosing γ .

- **Most of these records are lobe-dominated, one is not.** Three records lose 0.00 to 0.19 dB; sub-118/run-02, whose late deflection is over three times its lobe deflection, gains +0.59 dB non-causal and +0.49 dB buffered in its late window. Any corpus-level statement about γ is therefore a statement about the mix of records in the corpus.
- **Latency and coordinate are separable.** The one-beat-delay penalty is roughly constant across γ , so the streaming mode can be chosen independently of the coordinate.
- **Figure 1 is fine as it stands.** It is drawn at $\gamma = 0$ on a lobe-dominated record, where $\gamma = 0$ is the right choice; the warp would change it by -0.19 dB. If the warp enters the paper it should be as a formulation containing fixed lag as a special case, not as a performance claim.

Reproducing

```
python src/phase_swap_compare.py --out outputs/phase_swap.json
python src/phase_swap_report.py
```

The plain-text results are in `paper/phase_swap.txt`.